THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x \times \frac{5}{6} = \frac{5}{9}$, then $x \div \frac{5}{12}$ is: 1

(a) $1\frac{3}{5}$ **(b)** $3\frac{1}{5}$ **(c)** $1\frac{23}{30}$ **(d)** $\frac{4}{5}$

2. $\frac{7}{8} \div (\frac{7}{4} \div \frac{1}{4})$ equals: 1

(a) $\frac{1}{16}$ **(b)** 2 **(c)** $\frac{1}{8}$ **(d)** $\frac{1}{4}$

3. $\frac{2}{3}$ of a number is 14. Three times the number is: 1

(a) 63 **(b)** $63\frac{1}{2}$ **(c)** 126 **(d)** $31\frac{1}{2}$

4. A pipe fills $\frac{3}{8}$ of a tank in 9 minutes. The time to fill the whole tank (minutes) is: 1

(a) 24 **(b)** 12 **(c)** 48 **(d)** $24\frac{1}{2}$

5. $\frac{1}{4}$ of a wall is already painted. Then $\frac{4}{9}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{5}{24}$ **(c)** $\frac{77}{156}$ **(d)** $\frac{5}{6}$

6. Which has the greatest value? 1

(a) $6 \div \frac{2}{3}$ **(b)** $6 \div \frac{3}{2}$ **(c)** $6 \times \frac{3}{4}$ **(d)** $6 \times \frac{2}{3}$

7. $\frac{5}{12}$ of $2\frac{2}{5}$ is: 1

(a) $\frac{1}{2}$ **(b)** $2\frac{49}{60}$ **(c)** 2 **(d)** 1

8. If $x \times \frac{7}{9} = \frac{7}{6}$, then $x \div \frac{3}{1}$ is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{4}$ **(c)** 1 **(d)** $\frac{1}{2}$

9. **Assertion (A):** $1\frac{1}{2} \times \frac{2}{3} = 1$. 1
Reason (R): $1\frac{1}{2} = \frac{3}{2}$ and $\frac{3}{2} \times \frac{2}{3} = 1$.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The reciprocal of 0 is not defined. 1
Reason (R): The reciprocal of 1 is 1.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the fraction exactly halfway between $\frac{1}{6}$ and $\frac{5}{6}$, and verify that its distance from each is the same. **2**

12. Simplify $\left(\frac{2}{3}+\frac{1}{6}\right) \div \left(\frac{2}{3}-\frac{1}{6}\right) \times \frac{5}{7}$. **2**

13. After $\frac{5}{11}$ of a number is removed, 42 is left. Find the number and the part removed. **2**

14. Find the fraction exactly halfway between $\frac{3}{8}$ and $\frac{7}{12}$, and verify that its distance from each is the same. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Ravi alone can finish a job in 8 days and Sita alone in 24 days. What fraction of the job do they finish together in one day, and how many days do they take together? **3**

16. A cyclist rides at $\frac{33}{4}$ km per hour for 4 hours, then at $\frac{11}{2}$ km per hour for 6 hours. Find the total distance and the average speed for the whole journey. **3**

17. A pole of 17 m is cut into pieces of $1\frac{1}{5}$ m each. How many pieces are obtained and how much is left? **3**

OR

Ravi alone can finish a job in 9 days and Sita alone in 18 days. What fraction of the job do they finish together in one day, and how many days do they take together? **3**

18. A cyclist rides at $\frac{25}{2}$ km per hour for 6 hours, then at $\frac{35}{3}$ km per hour for 3 hours. Find the total distance and the average speed for the whole journey. **3**

OR

A pole of 22 m is cut into pieces of $2\frac{2}{3}$ m each. How many pieces are obtained and how much is left? **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A ribbon is $3\frac{1}{2}$ m long. Kiran cuts it into equal pieces. **5**

(i) Write $3\frac{1}{2}$ as an improper fraction.

(ii) Write the reciprocal of $\frac{7}{8}$.

(iii) How many pieces of $\frac{7}{8}$ m can be cut?

(iv) If each piece is $\frac{3}{4}$ m instead, how many full pieces are cut and what length is left over?

20. A farmer's land is 300 acres. He sells $\frac{2}{5}$ of it, gives $\frac{5}{9}$ of the remainder to his son and plants $\frac{3}{5}$ of what is left after that with sugarcane. Find each area and the area left unplanted. **5**

OR

Pipe A fills a tank in 6 hours, pipe B in 9 hours, and a drain empties the full tank in 18 hours. Find the fraction filled in 1 hour with all three open, and the time to fill the empty tank. **5**

END OF QUESTION PAPER — Total: 40 Marks